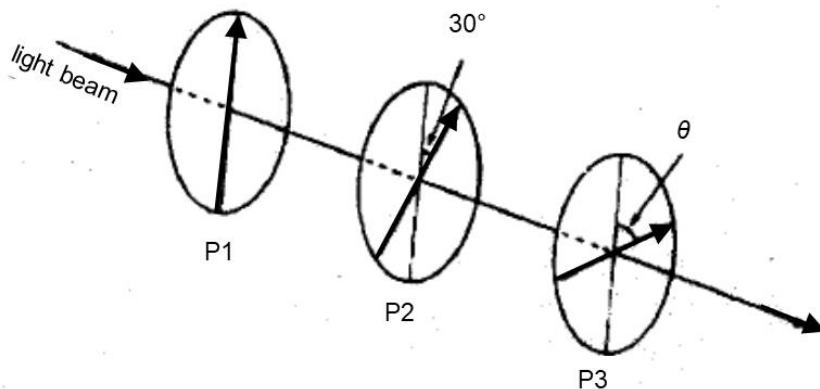


- 14 A beam of unpolarised light passes through three polaroids P_1 , P_2 and P_3 . The polarising axis of each polaroid is shown by an arrow. Polaroids P_1 and P_2 are fixed, with their polarising axes at 30° to each other and P_3 can be set with its polarising axis at a variable angle θ to that of P_1 .



For what values of θ do the emergent light have the lowest intensity?

- A $30^\circ, 120^\circ$
- B $60^\circ, 240^\circ$
- C $90^\circ, 270^\circ$
- D $120^\circ, 300^\circ$

Ans: **D**

To achieve minima for emergent light, P_3 has to be perpendicular to P_2 axis, i.e. $90^\circ + 30^\circ = 120^\circ$, and $180^\circ + 120^\circ = 300^\circ$

- 15 A point source of sound emits energy equally in all directions at a constant rate and a